

Process Mining and Formal Verification of the Volvo IT Incident Process

BPI Challenge 2013

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- An IT service desk records every state change of every incident.
- The process is *documented*. What was actually *executed*?
- Volvo IT Belgium, VINST system, 2010 to 2012.
- Three logs: incidents, open problems, closed problems.
- A standard benchmark, so the results are comparable with the literature.
- Nothing here is read off a dashboard by eye: every figure shown is produced by code and pinned by a regression test.

Log	Cases	Events	Activities	From	To
closed_problems	1,487	6,660	7	2006-01-11	2012-05-31
incidents	7,554	65,533	13	2010-03-31	2012-05-22
open_problems	819	2,351	5	2006-11-07	2012-06-15

Case and event counts reproduce the published figures exactly, the first check on the loader.

The problem logs are not variants of the incident process: median duration differs by an order of magnitude and their alphabet is smaller. The incident log is the object of study; the other two are used for comparison.

There is no activity column

The dataset records Status and Sub Status. What counts as an activity is the analyst's decision.

Status: 4 activities

Accepted, Queued, Completed, Unmatched

Small models. Waiting on a vendor and waiting on a user become indistinguishable.

Status+Sub Status: 13 activities

Accepted+In Progress, Accepted+Wait - User, ...

Three times larger models, lower precision, and the real process.

Treated as a parameter of the loader, not fixed silently. Both are reported.

Three discovery algorithms

Alpha Ordering relations from directly-follows pairs. Assumes a complete, noise-free log. Cannot represent short loops.

Heuristics Frequency-weighted, with a dependency threshold. Filters infrequent behaviour instead of modelling it. Robust to noise.

Inductive Recursive partitioning into a process tree, converted to a net. Guarantees soundness and, at zero noise, perfect fitness.

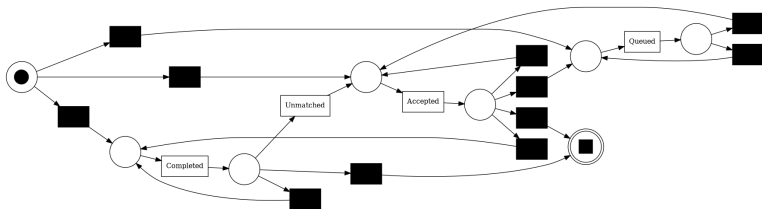
All three are run on the full incident log, under both labellings, from the same loaded bundle, so the six rows on the next slides differ only in the algorithm and the labelling.

Mode	Algorithm	Fitness	Precision	Gen.	Simpl.	Places	Trans.
status	alpha	0.119	0.400	0.734	1.000	2	4
status	heuristics	0.994	0.678	0.767	0.622	8	15
status	inductive	1.000	0.440	0.748	0.667	17	23
status+substatus	alpha	0.127	0.176	0.770	0.789	2	13
status+substatus	heuristics	0.949	0.385	0.634	0.471	27	69
status+substatus	inductive	1.000	0.234	0.783	0.634	57	78

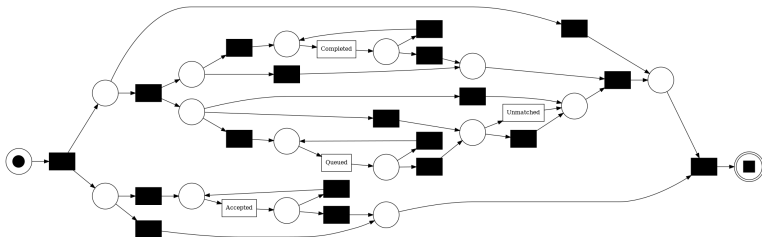
Four dimensions, and they trade off: fitness (can the model replay the log?), precision (does it forbid what never happened?), generalisation and simplicity. Inductive buys perfect fitness with permissiveness and the largest net. Heuristics gives the best balance. Alpha collapses.

Moving from 4 labels to 13 lowers precision for every algorithm and roughly triples the net: there is simply more distinct behaviour to explain.

What the two usable miners produce



Heuristics: 8 places, 15 transitions

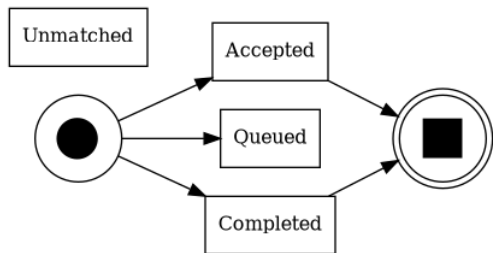


Inductive: 17 places, 23 transitions, plus silent routing

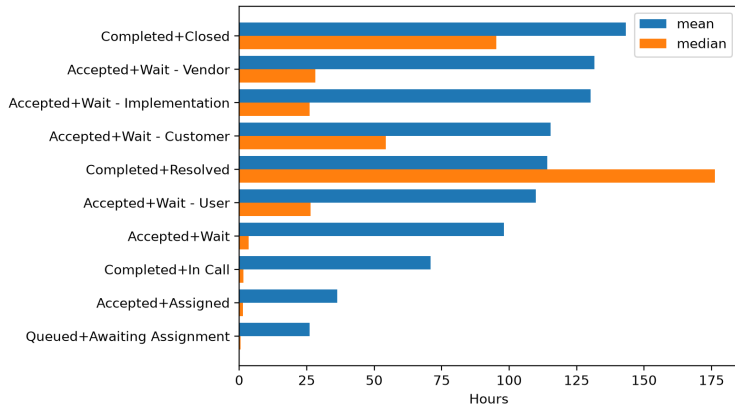
Why Alpha collapses

Fitness below 0.15; the net degenerates to two places. **This is not a bug.** 90% of cases contain a

length-one loop, an activity immediately repeated. Alpha infers contradictory ordering relations from a directly-follows pair (a, a) and provably cannot represent such loops. A known theoretical boundary, demonstrated on a real log.

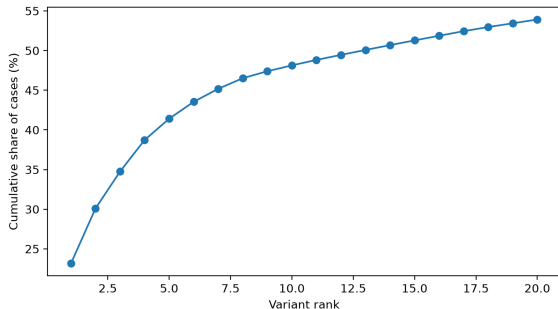


Where does the time go?



The Wait states dominate: delay is *external* to the service desk. Where mean and median diverge (Accepted+In Progress is minutes at the median, hours at the mean) work is normally picked up at once and occasionally stalls for months. Either figure alone misleads.

A weakly standardised process



Thousands of distinct variants against 7,554 cases. Most variants occur exactly

once. Consequence: any high-precision

model of this log would be overfitted. The low precision figures are the correct result.

Eight properties, each a real formula evaluated per case.

Termination	$F(\text{completed}_*)$
Resolution	$G(\text{in_progress} \rightarrow F \text{ completed}_*)$
No work after closure	$G(\text{closed} \rightarrow \neg F \text{ in_progress})$
Queued work picked up	$G(\text{queued} \rightarrow F \text{ in_progress})$
Formal closure	$G(\text{in_progress} \rightarrow F \text{ closed})$
Queueing precedes work	$(\neg \text{in_progress}) \cup \text{queued}$

The last two encode the *documented* procedure. They are expected to fail. The size of the gap is the result.

Property	Satisfied	Violated	Ratio
Termination	7,546	8	0.999
Resolution	7,545	9	0.999
No work after closure	7,435	119	0.984
Queued work is picked up	7,494	60	0.992
Waiting on the user resumes	7,550	4	0.999
No unmatched events	7,549	5	0.999
Formal closure	5,591	1,963	0.740
Queueing precedes work	1,168	6,386	0.155

Every case of the incident log, every property. No sampling.

Fewer than half of all events are the *first* occurrence of their activity within the case.

- Over half of all recorded work is repetition.
- Completion is near perfect; variability is moderate.
- Rework is the single component dragging the composite score down, and the largest available lever.

$(\neg in_progress) \cup queued$ holds for a small minority of cases.

- The documented procedure: queue the incident, then work it.
- What happens: work begins without the incident ever being queued.
- Established by *formal verification over every case*, not by sampling, not by eyeballing variants.

The packages

`logs` loading, schema detection, filters

`mining` discovery, conformance,
performance

`verification` temporal logic per trace

`analysis` score, anomalies, prediction

`ui`, reporting and scripts sit on top;
nothing depends on `ui`.

Consequences

- Every function takes a loaded log and returns a value: no module state, no cache
- Dashboard callbacks are plain functions, called directly by 211 tests
- Two browser sessions are independent
- Every table and figure on these slides is generated from code

It is one application, not a folder of scripts

Everything shown in this talk is reachable from a single Gradio dashboard, started with one command:

```
uv run python -m volvo.main
```

Eight tabs

Data load a log, choose the activity label

Preprocessing filters, deduplication

Discovery miner, threshold, Petri net

Conformance the four dimensions

LTL the shipped properties, or your own

Analytics waiting times, variants, score

Report the document, generated

Assistant questions in natural language

- **Change the labelling and rediscover.** The Status against Status+Sub Status comparison is two clicks, not two scripts.
- **Type a formula.** The LTL box accepts an arbitrary temporal-logic property over the activity alphabet and evaluates it against every case, returning violating cases as counterexamples. The eight shipped properties use that same code path.
- **Swap the miner and the threshold** and watch precision and net size move together.
- **Generate the report** and ask the assistant about the loaded log, with OpenAI, Gemini or a local Ollama model, or none at all.

State lives per browser session, so two people using it do not collide.

- Conformance is computed on a *sample*; alignment-based computation does not scale to the full log.
- Token-based replay is optimistic on unsound nets. Alignments agree on the sample.
- The composite score's weights are *stipulated*. Components are meaningful; the aggregate only comparatively.
- Unmatched+Unmatched is an export artefact, not work. Retained so its prevalence is visible.
- The analysis reflects what was *recorded*. Poor recording discipline looks like rework.

- ① **Rework dominates:** over half of recorded work is repetition.
- ② **Documented $\neq$ executed:** push-to-front proved formally over every case.
- ③ **Alpha is inapplicable here:** predicted by length-one loops, not an empirical surprise.

Next: the organisational attributes are already loaded. Handover analysis across support lines, push-to-front quantified by organisation, and incident versus problem as interacting processes.

Questions